IGCSE 0607 Practice Questions — E1.10 — Combined Questions & Markscheme

Ratio and proportion

Total Questions: 18 — **Total Marks:** 105 Marks

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- **E1.10** Ratio and proportion
  → **Paper 2 (Non-Calculator)** *(8 Qs)* · **Paper 4 (Calculator)** *(10 Qs)*

1 **(a)** Write 45 : 75 : 105 in its simplest form.

..... **[1]**

(b) Alice, Bob and Charlie share \$240 in the ratio of their ages. Alice is 8 years old, Bob is 10 years old and Charlie is 12 years old.
Work out how much Bob receives.

% ..... **[2]**

2 **(a)** Write the ratio 1.5 hours to 45 minutes in its simplest form.

..... **[1]**

(b) Alex, Ben and Carl share \$240 in the ratio 2 : 3 : 5.
Calculate how much more Carl receives than Alex.

% ..... **[2]**

3 **(a)** Write the ratio 450 g to 1.2 kg in its simplest form.

..... **[2]**

(b) Anna, Ben and Chloe share \$420 in the ratio 3 : 4 : 5.
Calculate the amount Ben receives.

\$ ..... **[2]**

4 Alex, Ben and Charlie share \$840 in the ratio $2 : 3 : 5$.

(a) Calculate the amount of money Ben receives.

\$ ..... **[2]**

(b) Express Charlie's share as a fraction of the total amount, giving your answer in its simplest form.

..... **[2]**

5 Alex, Ben and Chloe share \$120 in the ratio 3 : 4 : 5.

(a) Show that Chloe receives \$50.

[2]

(b) Work out the amount Alex receives.

\$ ..... **[2]**

(c) Write down the ratio of Ben's share to Chloe's share in its simplest form.

..... **[1]**

6 **(a)** Write the ratio 45 minutes to 2 hours in its simplest form.

..... **[2]**

(b) A piece of wood of length 3.6 m is cut into three pieces in the ratio 2 : 3 : 7.
Calculate the length, in cm, of the longest piece.

..... cm **[3]**

7 A sports club has m male members and f female members.
When 30 new male members join the club, the ratio of male members to female members becomes 3 : 1.
When, instead, 20 new female members join the original members, the ratio of male members to female members becomes 1 : 1.

(a) Write down two simultaneous equations in m and f.

.....

..... **[2]**

(b) Solve the equations to find the value of m and the value of f.

$m =$ ..... , $f =$ ..... **[3]**

(c) Find the original ratio of male members to female members in its simplest form.

..... **[1]**

8 A bag contains red balls, blue balls and green balls.
The ratio of red balls to blue balls is 3 : 4.
The ratio of blue balls to green balls is 5 : 2.

(a) Find the ratio of red balls to blue balls to green balls in its simplest form.

..... **[2]**

(b) There is a total of 129 balls in the bag.
Calculate the number of green balls.

..... **[2]**

(c) y blue balls and y green balls are removed from the bag.
The new ratio of blue balls to green balls is 3 : 1.
Work out the value of y.

$y =$ ..... **[3]**

9 A map is drawn to a scale of 1 : 50000.

(a) The length of a road on the map is 4.5 cm.
Calculate the actual length of the road in kilometres.

..... km **[2]**

(b) A lake has an actual area of 14.2 km^2.
Calculate the area of the lake on the map in cm^2.

..... cm^2 **[3]**

10 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two towns on the map is 8.4 cm. Calculate the actual distance between the two towns in kilometres.

..... km **[3]**

(b) A forest has an actual area of 35 square kilometres.
Calculate the area of the forest on the map in square centimetres.

..... cm^2 **[3]**

11 A map is drawn to a scale of 1 : 250 000.

(a) The distance between two cities on the map is 16.4 cm.
Calculate the actual distance between the two cities in kilometres.

..... km **[2]**

(b) The actual area of a lake is 35 km^2.
Calculate the area of the lake on the map in square centimetres.

..... cm^2 **[3]**

(c) Another map is drawn such that a length of 5 cm represents an actual distance of 20 km.
Write the scale of this map in the form 1 : n.

1 : ..... **[1]**

12 **(a)** A map has a scale of 1 : 50000.
The actual distance between two towns is 14.5 km.
Calculate the distance, in cm, between the two towns on the map.

..... cm **[2]**

(b) An area of a forest on the map is 12.4 cm^2.
Calculate the actual area of the forest in km^2.

..... km^2 **[2]**

(c) A recipe for 8 people requires 240 g of flour and 150 g of butter.
Marco wants to make the recipe for 12 people.
He has 400 g of flour and 200 g of butter.
Determine if Marco has enough ingredients.
Show your working.

.....

..... **[3]**

13 A map is drawn to a scale of $1 : 250\,000$.

(a) The distance between two cities on the map is 16.8 cm.
Calculate the actual distance between the cities, in km.

..... km **[2]**

(b) A lake has an actual area of 56 km^2.
Calculate the area of the lake on the map, in cm^2.

..... cm^2 **[2]**

(c) On a different map of the same region, the area of the same lake is 14 cm^2.
Find the scale of this new map in the form $1 : n$.

$1 :$ ..... **[3]**

14 **(a)** The ratio of adults to children in a theater is 5 : 3.
There are 84 more adults than children.
Calculate the total number of people in the theater.

..... **[3]**

(b) A map is drawn to a scale of 1 : 25000.

(i) Two towns are 8.5 cm apart on the map.
Calculate the actual distance between the towns in kilometres.

..... km **[2]**

(ii) A forest has an actual area of 3.5 km^2.
Calculate the area of the forest on the map in square centimetres.

..... cm^2 **[3]**

15 In a factory, the number of male workers to female workers is in the ratio 5 : 3.
The ratio of male workers who work full-time to male workers who work part-time is 3 : 2.
The ratio of female workers who work full-time to female workers who work part-time is 4 : 1.

(a) Find the ratio of the total number of full-time workers to the total number of part-time workers in the factory.
Give your answer in its simplest form.

..... **[4]**

(b) There are 42 more full-time male workers than part-time female workers.
Calculate the total number of workers in the factory.

..... **[3]**

16 In a factory, the number of workers on the day shift and the number of workers on the night shift are in the ratio 5 : 3.

(a) There are 42 more workers on the day shift than on the night shift.
Calculate the total number of workers in the factory.

..... **[3]**

(b) The ratio of male workers to female workers on the day shift is 3 : 4.
The ratio of male workers to female workers on the night shift is 2 : 7.
Calculate the overall ratio of male workers to female workers in the factory.
Give your answer in its simplest form.

..... **[4]**

17 **(a)** In a club, the ratio of the number of boys to the number of girls is 3 : 5.
When 4 boys leave and 8 girls join the club, the new ratio of boys to girls becomes 1 : 3.
Calculate the original number of boys in the club.

..... **[4]**

(b) The ratio of the mass of object A to object B is 3 : 4.
The ratio of the mass of object B to object C is 5 : 6.
The mass of object C is 45 kg more than the mass of object A.
Calculate the total mass of the three objects.

..... kg **[3]**

18 Three partners, Alex, Ben and Chris, share a total profit of $\$P$. The ratio of Alex's share to Ben's share is $3 : 4$. The ratio of Ben's share to Chris's share is $5 : 6$.

(a) Find the ratio of Alex's share to Ben's share to Chris's share in its simplest form.

..... **[2]**

(b) Chris receives \$16 200 more than Alex. Calculate the value of P.

\$ ..... **[3]**

(c) Alex invests his share of the profit in a bank account that pays compound interest at a rate of 4.2% per year.
Calculate the value of Alex's investment at the end of 5 years.
Give your answer correct to the nearest dollar.

\$ ..... **[3]**

1 **(a)** $3 : 5 : 7$ **[B1]**

(b) 80 **[B2]**

If 0 scored, M1 for $\frac{240}{8+10+12} \times 10$ **[M1]**

[Back to Question] *[Total Marks: 3]*

2 **(a)** $2 : 1$ **[B1]**

(b) $\frac{240}{2+3+5} \times (5-2)$ or $120 - 48$ **[M1]**

72 **[A1]**

[Back to Question] *[Total Marks: 3]*

3 **(a)** M1 for 1.2 kg = 1200 g or 450 g = 0.45 kg

$3 : 8$ **[A1]**

(b) M1 for $3 + 4 + 5$ or 12 seen

140 **[A1]**

[Back to Question] *[Total Marks: 4]*

4 **(a)** Total parts $= 2 + 3 + 5 = 10$ **[M1]**

Ben's share $= \frac{3}{10} \times 840 = 252$ **[A1]**

(b) $\frac{5}{10}$ **[M1]**

$\frac{1}{2}$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 **(a)** $3 + 4 + 5 = 12$ **[M1]**

$\frac{5}{12} \times 120 = 50$ **[A1]**

(b) $\frac{3}{12} \times 120$ or 3×10 **[M1]**

30 **[A1]**

(c) $4 : 5$ **[B1]**

[Back to Question] *[Total Marks: 5]*

6 **(a)** $45 : 120$ **[M1]**

$3 : 8$ **[A1]**

(b) $3.6 \times 100 = 360$ cm **[M1]**

$\dfrac{360}{2+3+7} \times 7$ **[M1]**

210 **[A1]**

[Back to Question] *[Total Marks: 5]*

7 **(a)** $\dfrac{m+30}{f} = 3$ or $m + 30 = 3f$ **[B1]**

$\dfrac{m}{f+20} = 1$ or $m = f + 20$ **[B1]**

(b) M1 for substituting to eliminate one variable, e.g., $f + 20 + 30 = 3f$ or $m + 30 = 3(m - 20)$

M1 for correctly simplifying their equation to $2f = 50$ or $2m = 90$

$m = 45$, $f = 25$ **[A1]**

(c) 9 : 5 **[B1]**

[Back to Question] *[Total Marks: 6]*

8 **(a)** 15 : 20 : 8 **[B2]**

If 0 scored, M1 for $3 \times 5 : 4 \times 5$ or $5 \times 4 : 2 \times 4$ **[M1]**

(b) 24 **[B2]**

If 0 scored, M1 for $\dfrac{129}{15+20+8} \times 8$ **[M1]**

(c) 6 **[B3]**

If 0 scored, M1 for $\dfrac{60-y}{24-y} = 3$ or $\dfrac{60-y}{24-y} = \dfrac{3}{1}$ **[M1]**

M1 for $60 - y = 72 - 3y$ (or equivalent linear equation) **[M1]**

[Back to Question] *[Total Marks: 7]*

9 **(a)** 4.5×50000 or 225000 seen **[M1]**

2.25 **[A1]**

(b) Area scale is 1 cm^2 : 0.25 km^2 or map area ratio $1^2 : 50000^2$ **[M1]**

$14.2 \div 0.25$ or $14.2 \times 10^{10} \div 50000^2$ **[M1]**

56.8 **[A1]**

[Back to Question] *[Total Marks: 5]*

10 **(a)** Actual distance $= 8.4 \times 250\,000 = 2\,100\,000$ cm **[M1]**

Distance in m $= 21\,000$ or distance in km $= \dfrac{2\,100\,000}{100\,000}$ **[M1]**

21 **[A1]**

(b) Area scale factor $= 1^2 : (250\,000)^2 = 1 : 6.25 \times 10^{10}$ **[M1]**

$35 \text{ km}^2 = 35 \times 10^{10} \text{ cm}^2$ **[M1]**

Map area $= \dfrac{35 \times 10^{10}}{6.25 \times 10^{10}} = 5.6$ **[A1]**

[Back to Question] *[Total Marks: 6]*

11 (a) 16.4×250000 or 4100000 **[M1]**

41 **[A1]**

(b) Scale factor for area is $(250000)^2$ or 1 km $= 10^5$ cm so 1 km$^2 = 10^{10}$ cm^2 **[M1]**

$\dfrac{35 \times 10^{10}}{250000^2}$ or $\dfrac{35}{2.5^2}$ **[M1]**

5.6 **[A1]**

(c) 400000 **[B1]**

[Back to Question] *[Total Marks: 6]*

12 (a) 29 **[B2]**

If 0 scored, M1 for $14.5 \times 1000 \times 100 \div 50000$ or equivalent **[M1]**

(b) 3.1 **[B2]**

If 0 scored, M1 for $12.4 \times \left(\dfrac{50000}{100000}\right)^2$ or 12.4×0.25 **[M1]**

(c) Calculates requires 360 g of flour and 225 g of butter **[B2]**

If B0 scored, M1 for $\dfrac{240}{8} \times 12$ or $\dfrac{150}{8} \times 12$ **[M1]**

No, because he needs 225 g of butter but only has 200 g **[B1]**

[Back to Question] *[Total Marks: 7]*

13 (a) $16.8 \times 250\,000 = 4\,200\,000$ cm **[M1]**

42 **[A1]**

(b) Scale of area $= 1^2 : (2.5)^2$ OR 1 cm^2 represents 6.25 km^2 **[M1]**

8.96 **[A1]**

(c) 14 cm^2 represents 56 $km^2 \implies$ 1 cm^2 represents 4 km^2 **[M1]**

Linear ratio $= \sqrt{1}$ cm represents $\sqrt{4}$ km $\implies$ 1 cm to 2 km **[M1]**

$200\,000$ **[A1]**

[Back to Question] *[Total Marks: 7]*

14 (a) M1 for $5 - 3 = 2$ parts

M1 for $84 \div 2$ or 42 seen

336 **[A1]**

(b) (i) M1 for 8.5×25000 or 212500 seen

2.125 **[A1]**

(ii) M1 for 1 cm represents 0.25 km or 1 km represents 4 cm

M1 for area scale factor 1 cm^2 represents 0.0625 km^2 or $3.5 \times 10^{10} \div 25000^2$

56 **[A1]**

[Back to Question] *[Total Marks: 8]*

15 (a) Let total workers be $8x$. Males $= 5x$, Females $= 3x$. **[M1]**

Full-time males $= \frac{3}{5} \times 5x = 3x$. Part-time males $= \frac{2}{5} \times 5x = 2x$. **[M1]**

Full-time females $= \frac{4}{5} \times 3x = 2.4x$. Part-time females $= \frac{1}{5} \times 3x = 0.6x$. **[M1]**

Total full-time $= 3x + 2.4x = 5.4x$. Total part-time $= 2x + 0.6x = 2.6x$. Ratio is $5.4 : 2.6 = 27 : 13$. **[A1]**

(b) Difference $= 3x - 0.6x = 2.4x$ **[M1]**

$2.4x = 42 \implies x = 17.5$ **[M1]**

Total workers $= 8x = 8 \times 17.5 = 140$ **[A1]**

[Back to Question] *[Total Marks: 7]*

16 (a) $5 - 3 = 2$ parts represent 42 workers **[M1]**

1 part is $\frac{42}{2} = 21$ **[M1]**

$8 \times 21 = 168$ **[A1]**

(b) Day shift workers: $5 \times 21 = 105$. Night shift workers: $3 \times 21 = 63$ **[M1]**

Day shift males: $\frac{3}{7} \times 105 = 45$. Day shift females: $105 - 45 = 60$ **[M1]**

Night shift males: $\frac{2}{9} \times 63 = 14$. Night shift females: $63 - 14 = 49$ **[M1]**

Total males $= 45 + 14 = 59$. Total females $= 60 + 49 = 109$. Ratio is 59 : 109 **[A1]**

[Back to Question] *[Total Marks: 7]*

17 (a) Let the number of boys be $3x$ and girls be $5x$.

Forming equation: $\frac{3x - 4}{5x + 8} = \frac{1}{3}$ [M2]

(or M1 for either $3x - 4$ or $5x + 8$ seen in a ratio)

$9x - 12 = 5x + 8$ **[M1]**

15 **[A1]**

(b) Common ratio $A : B : C = 15 : 20 : 24$ **[M1]**

Difference in parts $24 - 15 = 9$, so 9 parts $= 45$ kg

$\implies$ 1 part $= 5$ kg **[M1]**

Total mass $= (15 + 20 + 24) \times 5 = 295$ **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** Alex : Ben $= 15 : 20$ and Ben : Chris $= 20 : 24$ **[M1]**

$15 : 20 : 24$ **[A1]**

(b) Difference in parts $= 24 - 15 = 9$ parts $\implies$ 9 parts $= 16\,200$ **[M1]**

Total parts $= 15 + 20 + 24 = 59$ **[M1]**

$P = \dfrac{16\,200}{9} \times 59 = 106\,200$ **[A1]**

(c) Alex's share $= \dfrac{15}{59} \times 106\,200 = 27\,000$ **[B1]**

$27\,000 \times \left(1 + \dfrac{4.2}{100}\right)^5$ **[M1]**

$33\,167$ **[A1]**

[Back to Question] *[Total Marks: 8]*